8 Features of manufacturing industries

8.3 Modern manufacturing methods and systems when designing for manufacture for small, medium and large-scale production

Topic	What students need to learn:
8 Features of manufacturing industries	8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium and large scale production: *a) production scheduling and production logistics* *b) robotics in production – robots on fully-automated production and assembly lines/cells*

Topic	What students need to learn:
8 Features of manufacturing industries	*c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* *d) flexible manufacturing systems (FMS), modular/cell production systems* *e) lean manufacturing using just-in-time (JIT) systems* *f) standardised parts, bought-in components* *g) quick response manufacturing (QRM)* *h) data integration – product data management (PDM), enterprise resource planning (ERP) systems* *i) concurrent manufacturing.*

h) Data Integration

Product Data Management (PDM), Enterprise Resource Planning (ERP) Systems

Product Data Management (PDM)

- **Product Data Management (PDM)** is a centralised, secure information system used to synchronise all the data associated with a product.
- It prevents the data corruption issues sometimes encountered in Computer-Integrated Manufacture (CIM) systems that can lead to machinery malfunction.
- PDM data includes:
  - *Technical specifications, including measurements and the materials that are used*
  - *CAD and manual drawings - 3D models / CAD drawings*
  - *Bill of materials (BOM), including details of parts, sub-assemblies and raw material*
  - *Construction details relating to manufacturing processes*
  - *Costs*
  - *Details of suppliers involved*
  - *Any other relevant documents, such as photos of the final product*

Product Data Management (PDM)

- PDM is often part of a larger ***Product Lifecycle Management (PLM) system***, which tracks every aspect of products from initial ideas conceptions and marketing to withdrawal.

Examples of simplified PDM system sequence:

- *CAD design is modified until it is approved.*
- *The system co-ordinates stress analysis (FEA) of CAD design.*
- *Stress analysis is signed off within system.*
- *The system notifies manufacturing engineers that manufacture planning can begin.*
- *Tooling design is initiated.*
- *Data from completed stages is submitted to the system for review and team sign-off.*
- *Manufacture begins.*

Advantages and Disadvantages of PDM

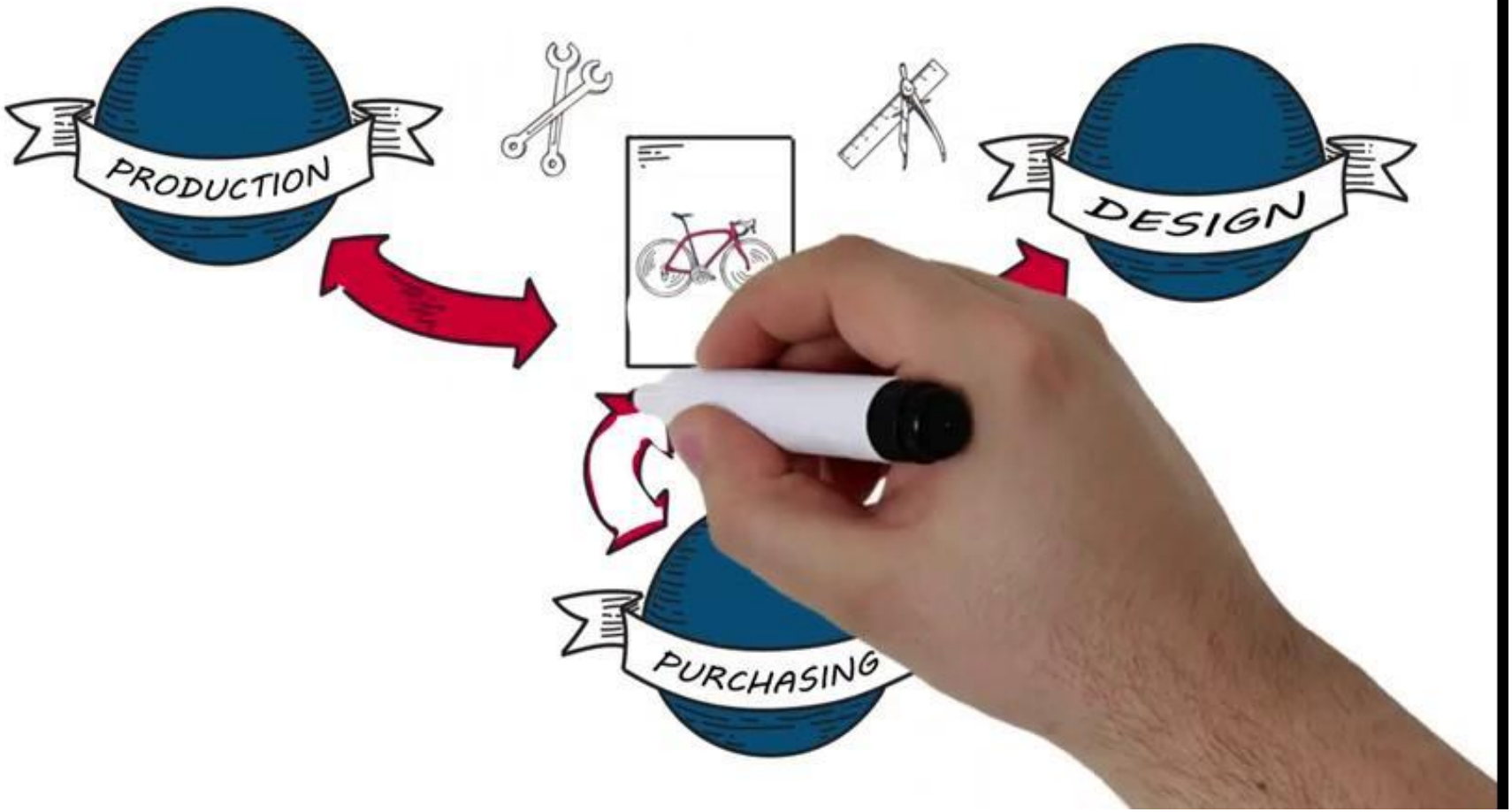
Advantages	Disadvantages
Products reach the market more quickly as reviews are carried out more efficiently	Searching for information in the system can sometimes be difficult if data input hasn't been carried out efficiently
Productivity is improved due to automatic tracking and management of data	Integrating information from different programs can be problematic
Control of the process is more efficient and secure, using only the most up-to-date data	Duplicate and large files can cause the system to slow down

Product Data Management (PDM)



Figure 3.15 *Product data management (PDM) systems enable efficient collaboration when planning production.*

Video: Why is Product Data Management important?



Enterprise Resource Planning (ERP) Systems

Enterprise Resource Planning (ERP) Systems is a *computer-based system* that ***integrates*** all the aspects of the development, manufacture and marketing of products in a single, unified database. Aspects that are typically incorporated include:

- *Finance*
- *Human Resources (HR)*
- *Manufacturing, including schedules, capacity, lead times and tracking*
- *Stock control and warehousing*
- *Customer orders, credit rating and order history*
- *Delivery schedules*
- *Business strategy and market research*
- *Quality*

Enterprise Resource Planning (ERP) Systems

Employees in the different departments all see the same information and can update it instantly.

When one department finishes with the order it is automatically sent via the ERP system to the next department.

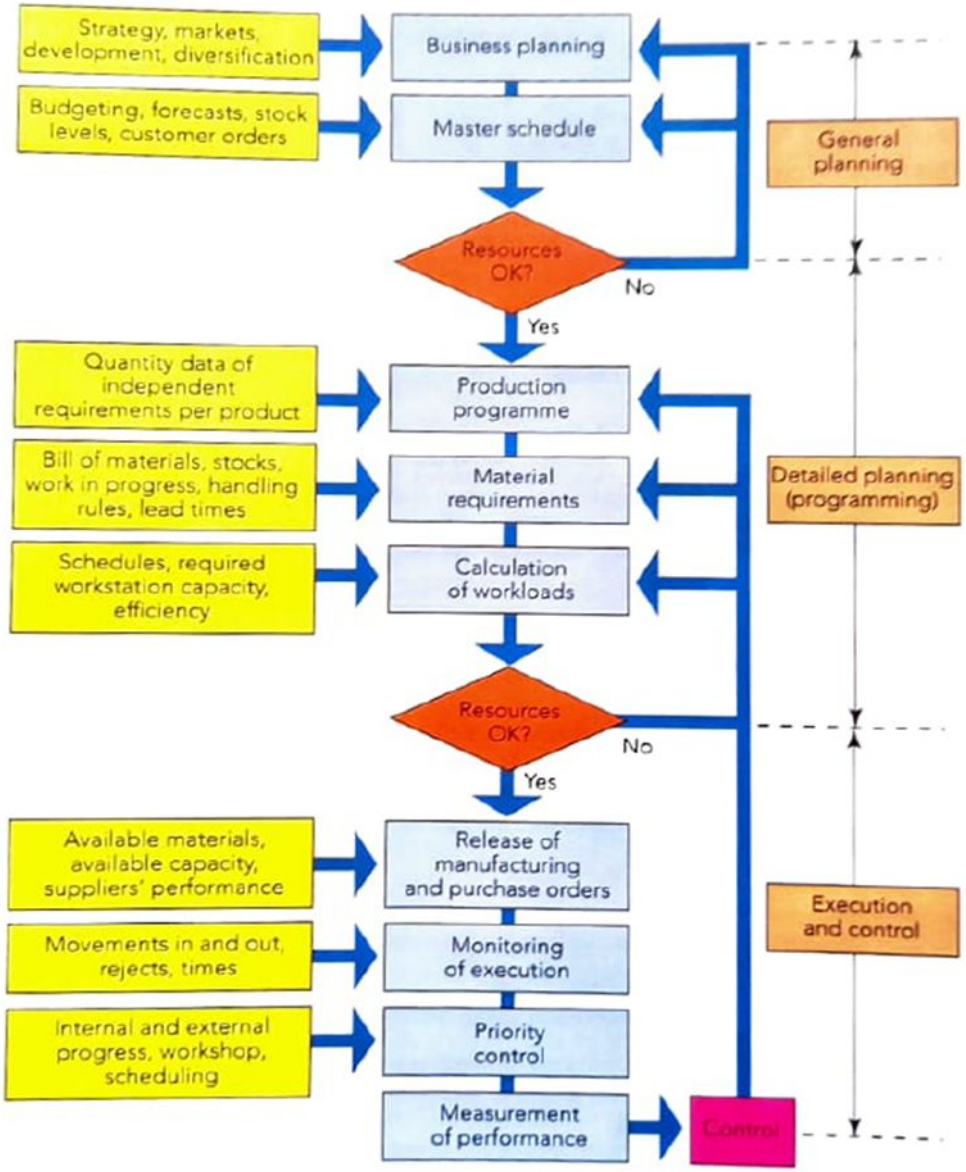
Therefore, any order can be easily tracked, and customers should receive delivery of their orders faster and without errors.

Advantages and Disadvantages of ERP

Advantages	Disadvantages
Productivity and efficiency are improved	Very expensive, particularly in initial, transitional stage
Operational costs are reduced as processes across departments can be merged	Complexity of implementation and staff training issues
It acts as an integrated information repository	The entire system can be compromised if just one element underperforms
Flexibility for a range of situations	
Helps to ensure that data is secure and that the company is complying with current regulations	

Enterprise Resource Planning (ERP) Systems

Enterprise resource planning (ERP) in a modern ***computer integrated manufacturing (CIM)*** system



Video: Enterprise Resource Planning (ERP) Systems

ERP Software integrates the different business functions to help them work better together…



For example:

- Communicating with suppliers
- Managing inventory deliveries
- Making payments to suppliers
- Handling quality problems

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Now Test Yourself:

1. Explain briefly why a manufacturing company might introduce an ***Enterprise Resource Planning (ERP)*** system.